

BIOGRAPHICAL SKETCH

NSF Common Form — Senior/Key Person

Identifying Information

Name: Davis, Issac D.

Persistent Identifier (PID/ORCID): N/A (not registered)

Position Title: Principal Investigator / Independent AI Safety Researcher

Organization and Location

Organization: ISSAC D DAVIS / SCBE-AETHERMOORE

Location: Port Angeles, WA, USA

A. Personal Statement

I am an independent AI safety researcher and sole developer of the SCBE-AETHERMOORE governance framework — a 14-layer composed-operator pipeline that enforces AI safety through hyperbolic geometry. My research centers on the insight that adversarial behavior in multi-agent systems can be made computationally infeasible by placing it in the exponentially-expensive periphery of Poincaré ball space. Governance becomes a geometric property of the system, not a post-hoc classifier.

I am self-taught in the relevant mathematics — Lyapunov stability analysis, control-barrier functions (CBF), port-Hamiltonian operator structure, and the Poincaré ball metric — which I independently derived through primary texts (Khalil; Marsden & Ratiu; Cannon) before encountering the formal field terminology. This approach — deriving first, naming second — has produced a framework that maps directly to DARPA MATHBAC's TA1 (multi-agent communication dynamics) and to the CLARA program's compositional AR+ML requirement.

I am uniquely qualified to execute this proposal because I have already built and benchmarked the system. Pre-submission evidence includes: 13/13 terminal-bench task parity (zero governance overhead), 172/173 adversarial classifications correct on the Anthropic Petri benchmark (99.42%), and a p95 latency of 0.1095ms for 240 RuntimeGate evaluations. The mathematical framework, benchmarks, and open-source codebase all existed before this proposal was written — this is evidence of genuine readiness, not proposal-generated claims.

B. Education and Training

* = required field. Entries follow Institution | Degree/Training | Field | Year format.

Institution / Organization	Degree / Training Type	Field	Year(s)
Self-directed study (primary texts: Khalil; Marsden & Ratiu;	Independent research training	Nonlinear control theory, hyperbolic geometry, port-	2020–present

Cannon; Brattka et al.)		Hamiltonian systems	
Self-directed study (primary texts: Sutton & Barto; Goodfellow et al.; Devlin et al.)	Independent research training	Machine learning, neural architecture, transformer systems	2019–present
Self-directed study (primary texts: Baier & Katoen; Clarke et al.)	Independent research training	Formal methods, model checking, logic-based AI	2021–present
Port Angeles, WA (professional practice)	Registered Entity (SAM.gov Active, UEI J4NXHM6N5F59, CAGE 1EXD5)	AI safety research and governance systems development	2023–present

C. Products

Most Relevant Products (up to 5)

1. Davis, I. (2025). The Six Tongues Protocol. KDP (Amazon). ASIN: B0GSSFQD9G. Timestamped prior art for the Langues Weighting System (phi-metric Sacred Tongue tokenization), which forms the L3–L4 weighted transform in the SCBE 14-layer pipeline.
2. Davis, I. (2024, filed). USPTO Provisional Patent Application 63/961,403. SCBE Composed-Operator Governance Framework — Poincaré ball safety geometry, 14-layer axiom-enforced pipeline, harmonic wall scoring $H(d, pd) = 1/(1 + \phi \cdot d_H + 2 \cdot pd)$.
3. Davis, I. (2024–present). SCBE-AETHERMOORE. Open-source repository (GitHub: [issdandavis/SCBE-AETHERMOORE](#)). 62+ TypeScript modules, 40+ Python modules implementing the full 14-layer pipeline, HYDRA orchestration, Juggling Scheduler, Red/Blue Arena, and the SacredTongues tokenizer. Apache 2.0 license.
4. Davis, I. (2024). [issdandavis/phdm-21d-embedding](#). HuggingFace Hub. Polyhedral Hamiltonian Defense Manifold embedding model; 21-dimensional canonical state lift for AI governance decision consistency.
5. Davis, I. (2025). [issdandavis/spiralverse-ai-federated-v1](#). HuggingFace Hub. Federated multi-agent governance model trained on SCBE-governed trajectories.

Other Significant Products (up to 5)

6. Davis, I. (2026). "Witnessed" (novel). KDP. ASIN: B0H257QJC2. Published work demonstrating applied narrative AI ethics concepts; fictional exploration of agent accountability and moral persistence themes.
7. Davis, I. (2026). "The Miracle Was the Memory" (novel). KDP. Published 2026-05-18. Companion work; explores memory and identity in AI-adjacent contexts.
8. Davis, I. (2026). SCBE Governance Evidence Brief (pre-submission benchmark package). Internal technical report; commit [f3fb4aa3c](#); includes 13/13 terminal-bench results, 172/173 Petri adversarial classification results, and 240-case RuntimeGate latency data.

D. Synergistic Activities

1. Open-source governance framework dissemination (2024–present). The SCBE-AETHERMOORE repository is publicly available under Apache 2.0, enabling external researchers and practitioners to examine, reproduce, and build on the 14-layer governance pipeline.
2. HuggingFace model releases (2024–2025). Released two governance models (phdm-21d-embedding; spiralverse-ai-federated-v1) to the public HuggingFace Hub under user issdandavis, providing the AI safety community with trained governance-aware embedding artifacts.
3. Prior art documentation and publication (2025). Published The Six Tongues Protocol (KDP, ASIN B0GSSFQD9G) as a timestamped technical record of the Sacred Tongues tokenization system, enabling independent verification of SCBE's pre-DARPA mathematical foundations.
4. DARPA DSO non-traditional proposer participation (2026). Submitted full proposals to DARPA-PA-25-07-02 (CLARA) and DARPA-PA-26-05 (MATHBAC) as an independent sole proprietor, demonstrating that foundational AI governance research is feasible outside traditional academic and large-corporate settings.
5. Collaboration with APEX Accelerator, Port Angeles, WA (2026–present). Receiving free government contracting guidance to build pathways for minority-owned small business access to federal research programs.

DRAFT — 2026-06-01. Prepared in accordance with NSF NSPM-33 Common Form (November 2023). Signature required for final submission. Certification: I certify that the information provided is current, accurate, and complete.